Failure to satisfy the sampling theorem (5.11) results in the overlapping of neighbouring spectra of an ideally sampled signal $S_{id}(\omega)$. This phenomenon is called **aliasing**. If we want to avoid overlapping, the continuous signal $s(t)$ must have the spectrum $S(\omega)$ bounded by a frequency filter of the lowpass type.

5.3 Recovery of continuous-time signal from its samples

The Fourier transform $S(\omega)$ of the original signal $s(t)$ can be obtained from the spectrum of an ideally sampled signal (unless aliasing occurs) by removing the lateral components using an ideal lowpass filter (Fig. 5.8) whose [illegible] frequency response is

$$H(\omega) = T \quad \text{for} \quad |\omega| \le \frac{\pi}{T} = \frac{\omega_s}{2}$$

$$H(\omega) = 0 \quad \text{for} \quad |\omega| > \frac{\omega_s}{2} \tag{5.12}$$

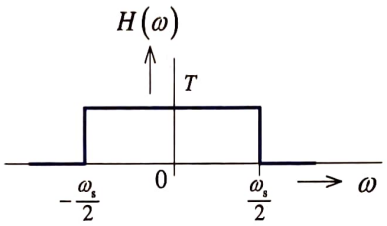


Figure 5.8: Frequency response of ideal reconstruction lowpass filter

The spectrum of the original continuous signal will be obtained as follows

$$S(\omega) = S_{id}(\omega)\, H(\omega) \tag{5.13}$$

The product in the spectral domain (5.13) is in the time domain replaced by the convolution

$$s(t) = s_{id}(t) * h(t) \tag{5.14}$$

where $h(t)$ is the impulse response of the ideal reconstruction lowpass filter. The impulse characteristic $h(t) \Leftrightarrow H(\omega)$ is

$$h(t) = \frac{1}{2\pi}\int_{-\infty}^{\infty} H(\omega)\, e^{j\omega t} d\omega = \frac{1}{2\pi}\int_{-\infty}^{-\frac{\omega_s}{2}} 0\, e^{j\omega t} d\omega + \frac{1}{2\pi}\int_{-\frac{\omega_s}{2}}^{\frac{\omega_s}{2}} T\, e^{j\omega t} d\omega + \frac{1}{2\pi}\int_{\frac{\omega_s}{2}}^{\infty} 0\, e^{j\omega t} d\omega$$

$$= \frac{T}{2\pi}\int_{-\frac{\omega_s}{2}}^{\frac{\omega_s}{2}} e^{j\omega t}\, d\omega = \frac{1}{\omega_s}\int_{-\frac{\omega_s}{2}}^{\frac{\omega_s}{2}} e^{j\omega t}\, d\omega \tag{5.15}$$

The calculation of the integral in equation (5.15) will be divided into two parts. For $t = 0$ it holds

$$h(0) = \frac{1}{\omega_s}\int_{-\frac{\omega_s}{2}}^{\frac{\omega_s}{2}} e^{j0t}\, d\omega = \frac{1}{\omega_s}\int_{-\frac{\omega_s}{2}}^{\frac{\omega_s}{2}} d\omega = \frac{1}{\omega_s}\,[\omega]_{-\frac{\omega_s}{2}}^{\frac{\omega_s}{2}} = 1 \tag{5.16}$$

For $t \neq 0$ it holds

$$h(t) = \frac{1}{\omega_s} \int_{-\frac{\omega_s}{2}}^{\frac{\omega_s}{2}} e^{j\omega t}\, d\omega = \frac{1}{\omega_s} \left[\frac{e^{j\omega t}}{jt} \right]_{-\frac{\omega_s}{2}}^{\frac{\omega_s}{2}} = \frac{1}{\omega_s} \frac{e^{j\frac{\omega_s}{2}t} - e^{-j\frac{\omega_s}{2}t}}{jt} = \frac{2}{\omega_s t} \sin\frac{\omega_s}{2} t = \frac{\sin\frac{\omega_s}{2} t}{\frac{\omega_s}{2} t} \tag{5.17}$$

Now we are able to solve the equation (5.14)

$$s(t) = s_{id}(t) * h(t) = \left\{ \sum_{n=-\infty}^{\infty} s[nT]\delta(t - nT) \right\} * \frac{\sin\left(\frac{\omega_s}{2} t\right)}{\frac{\omega_s}{2} t}$$

$$= \int_{-\infty}^{\infty} \sum_{n=-\infty}^{\infty} s[nT]\delta(\tau - nT) \frac{\sin\left(\frac{\omega_s}{2}(t-\tau)\right)}{\frac{\omega_s}{2}(t-\tau)} d\tau$$

$$= \sum_{n=-\infty}^{\infty} s[nT] \int_{-\infty}^{\infty} \frac{\sin\left(\frac{\omega_s}{2}(t-\tau)\right)}{\frac{\omega_s}{2}(t-\tau)} \delta(\tau - nT) d\tau = \sum_{n=-\infty}^{\infty} s[nT] \frac{\sin\left(\frac{\omega_s}{2}(t-nT)\right)}{\frac{\omega_s}{2}(t-nT)}$$

where we have again exploited the sampling property of the Dirac impulse. The result is

$$s(t) = \sum_{n=-\infty}^{\infty} s[nT] \frac{\sin\left(\frac{\omega_s}{2}(t-nT)\right)}{\frac{\omega_s}{2}(t-nT)} \tag{5.18}$$

In Fig. 5.9 we see how spaces between discrete samples are filled and the original signal $s(t)$ is recovered.

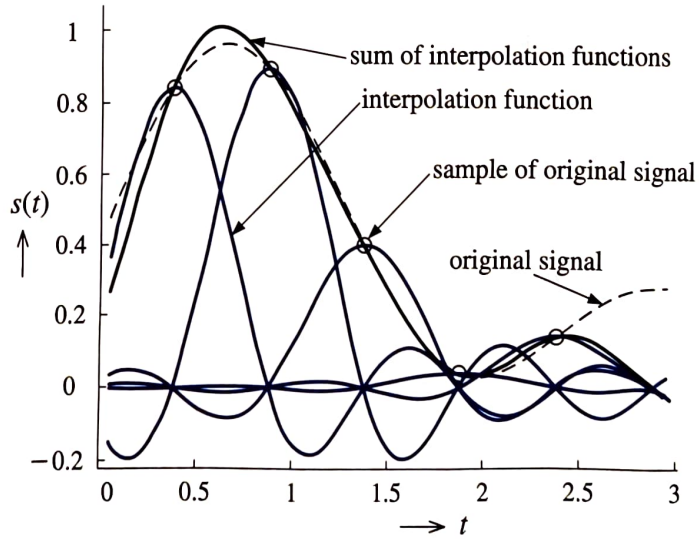


Figure 5.9: Recovering signal $s(t)$ from its samples $s[nT]$ using the ideal reconstruction lowpass filter in Fig. 5.8

It is obvious that the ideal lowpass filter (5.12) is a non-causal system. In practice, we must use a realizable lowpass filter (e.g. *RC* circuit in Fig. 4.5), and in that case the sampling theorem (5.11) can have the form

$$f_s > (\text{from } 2.35 \text{ to } 4) f_{max} \tag{5.19}$$

We give as an example the sampling frequency for a system of pulse-code modulation (PCM). In European countries the telephone band is defined in the range from 300 Hz up to 3400 Hz. By the sampling theorem it should hold that

$$f_s > 6800 \text{ Hz} \tag{5.20}$$

A sampling frequency of 8 kHz is chosen, i.e. it holds

$$f_s = \frac{8000}{3400} f_{max} = 2.353\, f_{max}$$

5.4 Sampling of band-limited signals

The band-limited continuous-time signal is a signal whose spectrum in the region of positive angular frequencies lies between the frequencies $\omega_1 < \omega_2$ while the bandwidth $B = \omega_2 - \omega_1$ is multiply smaller than the distance of the frequency ω_1 from the origin $\omega = 0$. An example of a band-limited signal can be seen in Fig. 5.10.

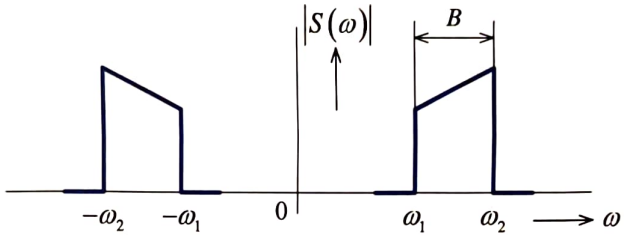


Figure 5.10: Band-limited continuous signal

The sampling conditions for a band-limited signal need not be as strict as they are given in the sampling theorem (5.11). For aliasing not to occur, the conditions of the following inequalities must be satisfied

$$\omega_1 > (n-1)\frac{\omega_s}{2} \quad \text{and} \quad \omega_2 < n\frac{\omega_s}{2}, \quad n \in \mathbb{N} \tag{5.21}$$

If we combine the two conditions, we obtain

$$\frac{2\omega_2}{n} < \omega_s < \frac{2\omega_1}{n-1}, \quad \frac{2f_2}{n} < f_s < \frac{2f_1}{n-1}, \quad 2 < n < \frac{f_2}{f_2 - f_1} \tag{5.22}$$

To better understand the problem, we set concrete values. Let there be a band-limited signal whose band lies between the frequencies $f_1 = 60$ kHz and $f_2 = 80$ kHz (defined for positive frequencies). By equations (5.22) it holds

$$2 < n < \frac{8 \cdot 10^4}{8 \cdot 10^4 - 6 \cdot 10^4} = 4 \quad \Rightarrow \quad n = 3$$

By using inequalities (5.22) we can for $n = 3$ write

$$\frac{16 \cdot 10^4}{3} < f_s < \frac{12 \cdot 10^4}{2} \quad \Rightarrow \quad 53.33\bar{3} \text{ kHz} < f_s < 60 \text{ kHz} \tag{5.23}$$

Fig. 5.10 gives two options of choosing the sampling frequency. In the first case a) the choice is $f_s = 55$ kHz, i.e. in compliance with the condition (5.23), the neighbouring spectra do not overlap, which means there is no aliasing. In the second case b) the condition for sampling band-limited signals (5.23) is not satisfied any more, i.e. the frequency chosen is $f_s = 50$ kHz, and we see that the neighbouring spectra start overlapping and aliasing occurs. Then it is no longer possible to obtain the original continuous signal.

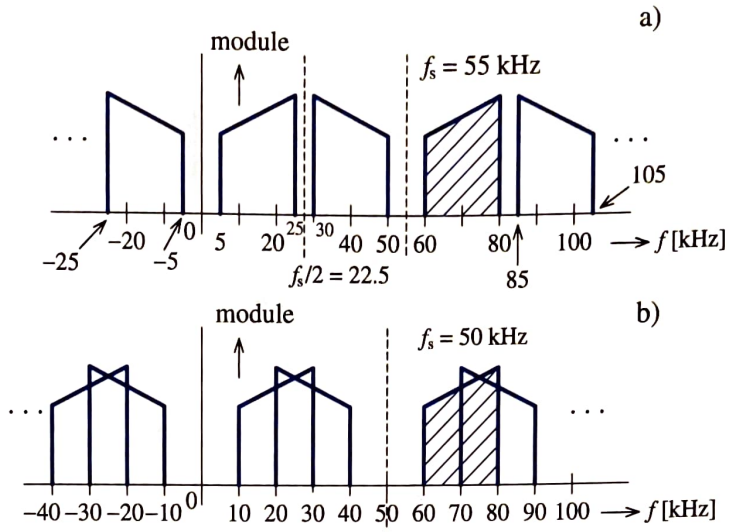


Figure 5.11: Example of sampling a band-limited signal (the baseband is shown hatched): a) $f_s = 55$ kHz, the repeated bands do not overlap and aliasing does not occur, b) $f_s = 50$ kHz, the case when aliasing occurs

5.5 First-order sampling

The first-order sampling is a procedure when short sections (Fig. 5.4) are cut out of the signal $s(t)$. This procedure can be expressed mathematically as follows

$$s_1(t) = s(t) \cdot u(t) \tag{5.24}$$

$$u(t) = \sum_{i=-\infty}^{\infty} u_0(t - iT), \qquad u_0(t) = A \quad \text{for} \quad |t| \leq \frac{\tau}{2} \tag{5.25}$$

The signal $u(t)$ is a periodic signal and it can be seen in Fig. 5.12.

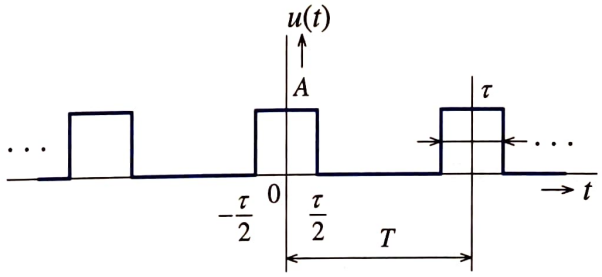


Figure 5.12: Sequence of rectangular impulses

We will expand the signal $u(t)$ into the Fourier series

$$u(t) = \sum_{k=-\infty}^{\infty} c_k \, e^{jk\frac{2\pi}{T}t}, \quad \omega_s = \frac{2\pi}{T} \tag{5.26}$$

The calculation of the Fourier series coefficients c_k will be divided into calculating for $k \neq 0$ and for $k = 0$

For $k \neq 0$ it will hold

$$c_k = \frac{1}{T}\int_{-\frac{T}{2}}^{\frac{T}{2}} u(t)\,\mathrm{e}^{-\mathrm{j}k\frac{2\pi}{T}t}\mathrm{d}t = \frac{1}{T}\int_{-\frac{T}{2}}^{-\frac{\tau}{2}} 0\,\mathrm{e}^{-\mathrm{j}k\frac{2\pi}{T}t}\mathrm{d}t + \frac{1}{T}\int_{-\frac{\tau}{2}}^{\frac{\tau}{2}} A\,\mathrm{e}^{-\mathrm{j}k\frac{2\pi}{T}t}\mathrm{d}t + \frac{1}{T}\int_{\frac{\tau}{2}}^{\frac{T}{2}} 0\,\mathrm{e}^{-\mathrm{j}k\frac{2\pi}{T}t}\mathrm{d}t =$$

$$= \frac{A}{T}\int_{-\frac{\tau}{2}}^{\frac{\tau}{2}} \mathrm{e}^{-\mathrm{j}k\frac{2\pi}{T}t}\mathrm{d}t = \frac{A}{T}\left[\frac{\mathrm{e}^{-\mathrm{j}k\frac{2\pi}{T}t}}{-\mathrm{j}k\frac{2\pi}{T}}\right]_{-\frac{\tau}{2}}^{\frac{\tau}{2}} = \frac{A}{k\pi}\cdot\frac{\mathrm{e}^{\mathrm{j}k\pi\frac{\tau}{T}} - \mathrm{e}^{-\mathrm{j}k\pi\frac{\tau}{T}}}{2\mathrm{j}} = \frac{A\tau}{T}\cdot\frac{\sin\left(k\pi\frac{\tau}{T}\right)}{k\pi\frac{\tau}{T}} \tag{5.27}$$

For $k = 0$ it holds

$$c_0 = \frac{1}{T}\int_{-\frac{\tau}{2}}^{\frac{\tau}{2}} A\,\mathrm{e}^{-\mathrm{j}0\cdot\frac{2\pi}{T}t}\mathrm{d}t = \frac{A\tau}{T} \tag{5.28}$$

Based on the results (5.27) and (5.28) we can write

$$c_k = \frac{A\tau}{T}\cdot\frac{\sin\left(k\pi\frac{\tau}{T}\right)}{k\pi\frac{\tau}{T}}, \quad k \in \mathbb{Z}, \quad k \neq 0 \tag{5.29}$$

By substituting into equation (5.24) we obtain the result

$$s_1(t) = s(t)\,u(t) = s(t)\sum_{k=-\infty}^{\infty} c_k\,\mathrm{e}^{\mathrm{j}k\frac{2\pi}{T}t} = s(t)\frac{A\tau}{T}\sum_{k=-\infty}^{\infty}\frac{\sin\left(k\pi\frac{\tau}{T}\right)}{k\pi\frac{\tau}{T}}\cdot\mathrm{e}^{\mathrm{j}k\frac{2\pi}{T}t} \tag{5.30}$$

If $S_1(\omega)$ is the Fourier transform of the signal $s_1(t)$, then it can be written

$$S_1(\omega) = \mathcal{F}\{s_1(t)\} = \mathcal{F}\left\{s(t)\sum_{k=-\infty}^{\infty} c_k\,\mathrm{e}^{\mathrm{j}k\frac{2\pi}{T}t}\right\} = \sum_{k=-\infty}^{\infty} c_k\,\mathcal{F}\left\{s(t)\mathrm{e}^{\mathrm{j}k\frac{2\pi}{T}t}\right\}$$

$$= \sum_{k=-\infty}^{\infty} c_k\,S\left(\omega - k\frac{2\pi}{T}\right) = \sum_{k=-\infty}^{\infty} c_k\,S(\omega - k\omega_s) \tag{5.31}$$

In the calculation we have exploited the following property of the Fourier transform

$$s(t)\mathrm{e}^{\mathrm{j}\Omega t} \Leftrightarrow S(\omega - \Omega), \qquad s(t) \Leftrightarrow S(\omega)$$

If the impulses $u_0(t)$ change to the Dirac impulses, first-order sampling will change to ideal sampling, the difference being that in relation (5.31) the Fourier series coefficients will be

$$c_k = \frac{1}{T}$$

5.6 Second-order sampling

In the second-order sampling the signal $s(t)$ is replaced by a sequence of rectangular impulses of the same shape but of a height that is proportional to the value of the signal $s(t)$ at the moment of sampling. Second-order sampling can be expressed mathematically as follows

$$s_2(t) = \sum_{i=-\infty}^{\infty} s[iT]\, u_0(t - iT) \tag{5.32}$$

where the rectangular impulse $u_0(t)$ is defined by (5.25). We will again employ the sampling property of the Dirac impulse and obtain

$$u_0(t - iT) = \delta(t - iT) * u_0(t) \tag{5.33}$$

Substituting equation (5.33) into equation (5.32) will yield

$$s_2(t) = \sum_{i=-\infty}^{\infty} s[iT]\, u_0(t - iT) = \sum_{i=-\infty}^{\infty} s[iT]\, [\delta(t - iT) * u_0(t)]$$

$$= \underbrace{\sum_{i=-\infty}^{\infty} s[iT]\, [\delta(t - iT)]}_{\text{ideal sampling}} * u_0(t) \tag{5.34}$$

The Fourier transform $S_2(\omega)$ will be obtained via performing the Fourier transform of $s_2(t)$

$$S_2(\omega) = \mathcal{F}\{s_2(t)\} = \mathcal{F}\left\{\sum_{i=-\infty}^{\infty} s[iT]\, [\delta(t - iT)]\right\} \cdot U_0(\omega)$$

$$= \left[\frac{1}{T}\sum_{k=-\infty}^{\infty} S(\omega - k\omega_s)\right] \cdot U_0(\omega) \tag{5.35}$$

The Fourier transform $U_0(\omega)$ will again be obtained by calculating the Fourier transform of the impulse $u_0(t)$, which is defined by the relation (5.25)

$$U_0(\omega) = \mathcal{F}\{u_0(t)\} = \int_{-\infty}^{\infty} u_0(t)\, \mathrm{e}^{-\mathrm{j}\omega t}\mathrm{d}t = A\int_{-\frac{\tau}{2}}^{\frac{\tau}{2}} \mathrm{e}^{-\mathrm{j}\omega t}\mathrm{d}t = A\tau\frac{\sin\left(\omega\frac{\tau}{2}\right)}{\omega\frac{\tau}{2}} \tag{5.36}$$

Substituting the result (5.36) into relation (5.35) gives

$$S_2(\omega) = \underbrace{\left[\frac{1}{T}\sum_{k=-\infty}^{\infty} S(\omega - k\omega_s)\right]}_{\text{ideal sampling}} \overbrace{A\tau\frac{\sin\left(\omega\frac{\tau}{2}\right)}{\omega\frac{\tau}{2}}}^{\text{distorting term}} \tag{5.37}$$

Fig. 5.13 gives a comparison of the module of the Fourier transform of the continuous-time signal $S(\omega)$ and the other Fourier transforms for the individual types of sampling, i.e. ideal, first-order and second-order sampling. If we choose spectrum $S(\omega)$ that is similar to the frequency response of the ideal lowpass filter, then we can show for the individual types of sampling how the type of sampling affects this spectrum. Ideal sampling will make the spectrum to be repeated at the multiples of the sampling frequency and its module will be multiplied by the coefficient $1/T$. Using the lowpass filter (5.12), the original spectrum $S(\omega)$ can be fully recovered. A similar situation arises in the case of first-order sampling, where subsequent to sampling, the modules of repeated spectra of the original signal are multiplied by the Fourier series coefficients of the expansion of the sequence of rectangular impulses. The situation after second-order sampling is, however, quite different because the spectrum of an ideally sampled signal is here distorted by the spectrum of the rectangular impulse and it is no longer possible to use the lowpass filter and ideally recover the spectrum of the original signal $s(t)$. Unfortunately, this type of sampling is the most frequent to occur in practice.

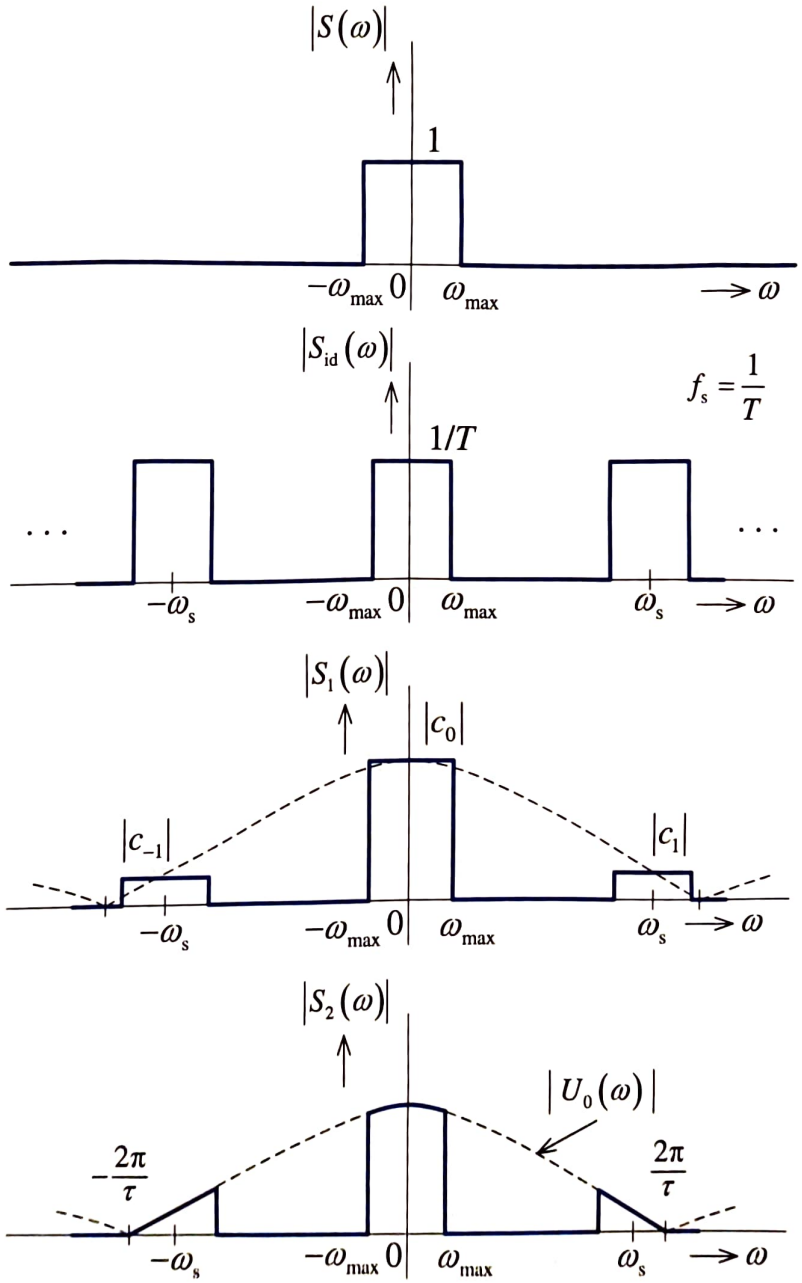


Figure 5.13: a) Module of the Fourier transform of continuous signal $|S(\omega)|$, b) module of the Fourier transform $|S_{id}(\omega)|$ for ideal sampling, c) module of the Fourier transform $|S_1(\omega)|$ for first-order sampling, and d) module of the Fourier transform $|S_2(\omega)|$ for second-order sampling

5.7 Amplitude quantization

As indicated at the beginning of this Section, after the sampling operation it is necessary to quantize the continuous-time signal samples and represent them in a suitable digital code. This will yield a digital signal. We will briefly return to the quantization operation. Second-order sampling is the most frequently used type of sampling in practice, which in fact means performing an **amplitude quantization**. Let us demonstrate the procedure of amplitude quantization for a harmonic signal with amplitude C_1 and angular frequency ω_1.

$$s(t) = C_1 \sin(\omega_1 t) \tag{5.38}$$

The waveform of one period of this harmonic signal prior and subsequent to amplitude quantization can be seen in Fig. 5.14. The result of amplitude quantization is given by the full stepped line $s_Q(t)$, with the original sinusoid signal $s(t)$ dashed.

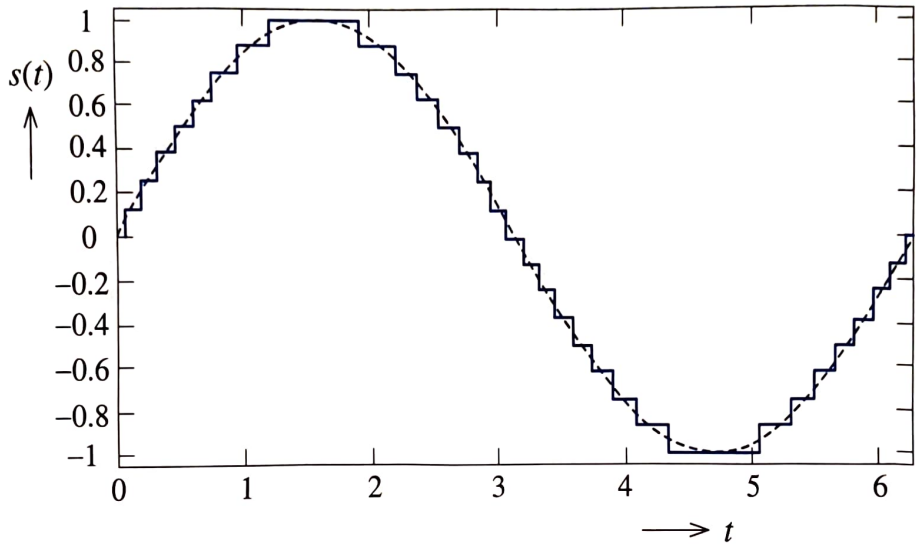


Figure 5.14: The waveform of one sinusoid period before (dashed black) and after amplitude quantization (blue)

The signal $s(t)$ is quantized by being rounded to the values iQ, $i = 1, 2, \ldots, K$, where Q is the quantization step and $K \in \mathbb{N}$. The effect of quantization can be looked at in several ways. We either introduce non-linearities into the processing, which are represented by non-linear characteristics (Fig. 5.15) or we maintain the quantization model linear and add sources of quantization noise. Introducing non-linearity into the model has a similar effect as in electric circuits. The principle of superposition does not hold, a shift in energy can occur in the spectrum, etc. Thus in the course of quantization, some information gets lost. The number of quantization levels is finite and it may happen that the signal $s(t)$ exceeds the limit values of the set of values being quantized, i.e. it is smaller than the value of the quantization step or, vice versa, it exceeds the maximum value KQ, K being the number of all quantization levels. This brings further potential losses of information during the amplitude quantization of signals, i.e. the losses are due to the amplitude of the signal being quantized.

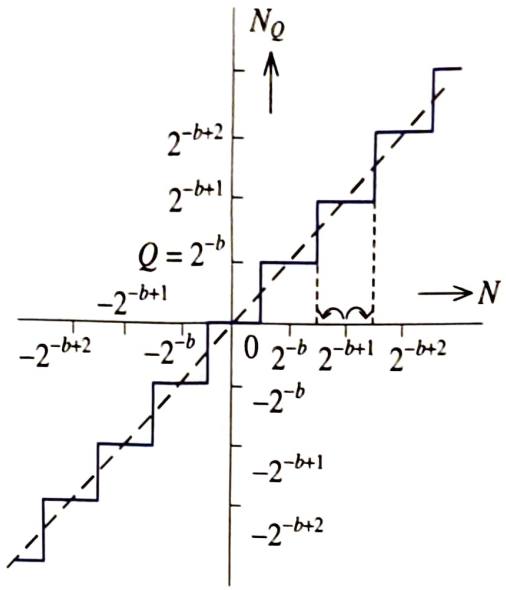


Figure 5.15: Transformation characteristic giving the relationship between non-quantized numbers N and quantized numbers N_Q (this is a rounding operation for all the three representations of fixed-point numbers)

The quantization noise is mostly of the sawtooth waveform. Its effective value can be calculated as the effective value of a periodic sawtooth signal, with a precision that is considered satisfactory in practice. The effective value of quantization noise is then given by the relation [RAB-75]

$$R_{\text{ef}} = \frac{Q}{\sqrt{12}} \tag{5.39}$$

The degree of damage to the original signal due to quantization can be expressed, for example, by the quantization distortion coefficient k, which is defined as a ratio of the effective value R_{ef} of the quantization noise (interference signal) $r(t)$ and the effective value S_{ef} of the useful signal. Quantization noise is defined as the difference

$$r(t) = s_Q(t) - s(t) \tag{5.40}$$

The distortion coefficient k for a harmonic signal (5.38) is given by the ratio of the effective values of the signals $r(t)$ and $s(t)$

$$k = \frac{R_{\text{ef}}}{S_{\text{ef}}} = \frac{\frac{Q}{\sqrt{12}}}{\frac{C_1}{\sqrt{2}}} = \frac{Q}{C_1\sqrt{6}}$$

Further information can be found, for example, in [RAB-75], [ROB-87], etc.